

Keep the observation separate from the chemical inference.

See the idea

A smell, a pop test or gas evolution is evidence. The name of the gas is the inference supported by that evidence.

Write the observation first. Then state the chemical claim it supports. The claim must not say more than the recorded observation or test can support.

The four-step reasoning route

- Copy the observation or test result without interpreting it.
- Write the possible chemical inference separately.
- Link the evidence to the inference.
- Check that the inference is no stronger than the evidence.

Worked example structure

Prompt: A gas is evolved and a test result is recorded. Separate the observation from the chemical inference before naming the conclusion.

OBSERVATION / TEST —————→ INFERENCE

The observation is copied exactly; the gas identity is written only after the supporting test is stated.

Guided attempt

Separate the supplied observation/test from the chemical inference and justify the link. Use the four-step routine shown in the lesson.

Faded attempt

Separate the supplied observation/test from the chemical inference and justify the link. Choose the second step yourself after writing the first evidence line.

Independent attempt

Separate the supplied observation/test from the chemical inference and justify the link. Use no procedural cue beyond the task and the representation.

Core1A ownership

- Teaches the distinction and the reasoning route.
- Chooses the governed representation.
- Owns guided/faded/independent teaching progression.